

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 10: Token Ring (IEEE 802.5) & LAN Topologies

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS TOKEN RING (IEEE 802.5)?

Token Ring is a collision-free Local Area Network (LAN) technology developed by IBM and standardized under IEEE 802.5. Unlike Ethernet, where computers compete to transmit, Token Ring passes a special small frame called a **Token** in a continuous logical circle.

Easy Everyday Example A meeting where only the person holding the microphone is allowed to speak. When finished, they hand the microphone to the next person. Because there is only 1 microphone, two people never talk over each other!

2. HOW TOKEN RING WORKS (STEP-BY-STEP)

Token Passing Process

- A small 3-byte frame called the **Free Token** circulates around the ring continuously.
- A station wanting to send data must wait until it catches the free token.
- It flips the token state from "Free" to "Busy", attaches its data frame behind it, and transmits it down the ring.

Destination & Release

- Every station checks the frame's destination address. Intermediate stations act as repeaters and forward it along.
- The intended receiver copies the data, marks the frame as "Copied/Received", and passes it back onto the ring.
- When the frame completes the loop and returns to the sender, the sender strips the data and releases a **Free Token** for the next station.

3. PHYSICAL STAR VS. LOGICAL RING (MAU)

In practice, Token Ring networks were not wired as a loose cable circle from room to room. Instead, devices connect via cables to a central unit called a **MAU (Multistation Access Unit)**.

- **Physical Topology:** Star (all cables plug into the central MAU box).
- **Logical Topology:** Ring (internally, the MAU routes the token from port to port in a closed circular loop).
- **Advantage:** If one workstation is unplugged or broken, relay switches inside the MAU automatically bypass the broken link without breaking the entire ring.

4. SUMMARY COMPARISON: ETHERNET (CSMA/CD) VS. TOKEN RING

Feature	Ethernet (IEEE 802.3)	Token Ring (IEEE 802.5)
Access Method	Contention-based (CSMA/CD)	Deterministic (Token Passing)
Collisions	Possible (in shared half-duplex)	Zero Collisions (guaranteed)
Fairness / Priority	Random; heavy loads cause delays	Fair round-robin; supports priority levels
Implementation Cost	Very inexpensive, simple hardware	Expensive, specialized MAU interfaces
Market Outcome	Dominates 100% of modern wired LANs	Replaced by Fast/Gigabit switched Ethernet

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why are collisions impossible in a Token Ring network?

Ans: Because there is only one active Token circulating on the ring at any time, and only the device currently holding that Token is permitted to transmit.

Q2: Why did Ethernet eventually beat Token Ring in the industry?

Ans: Ethernet hardware (NIC cards, cables, switches) was dramatically cheaper, easier to troubleshoot, and scaled faster into 100 Mbps, 1 Gbps, and 10 Gbps speeds.